LOTQ-0047 / STRATEGY DEVELOPMENT / BEGINNER / FIELD GUIDE

Signal Research and Strategy Ideation

Turn market observations into falsifiable hypotheses and traceable signal specifications.

Library of the Quant

Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0047
CATEGORY	Strategy Development
DIFFICULTY	Beginner
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USE	Study, lookup, and research-system reference

Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

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VISUAL LABS AND REVIEW

How to use this manual

BEGINNER

1. START WITH THE DECISION

Name the action, information cutoff, universe, horizon, benchmark, and rejection rule before opening the data. This prevents a good-looking statistic from becoming the objective after the fact.

2. WRITE THE MECHANISM

Identify the actor, incentive or constraint, transmission path, price response, decay, capacity limit, and evidence that would contradict the story. A narrative without distinct predictions is not a mechanism.

3. SEPARATE EXPLORATION AND CONFIRMATION

Explore distributions, lags, cohorts, and response shapes while logging every material choice. Lock the chosen hypothesis and preserve independent evidence for confirmation or replication.

4. DIAGNOSE BEFORE BACKTESTING

Use information coefficient, ordered portfolios, baseline controls, decay, parameter surfaces, negative controls, and concentration analysis to understand what the signal actually measures.

5. CARRY THE CLOCK AND COST

Every input needs an availability time and every action needs an executable price. Model delay, turnover, spread, impact, borrow, and capacity before interpreting gross prediction as a strategy.

6. CONNECT EVIDENCE TO CONTROLS

A promoted candidate requires pinned lineage, search accounting, independent challenge, parity tests, pilot limits, monitors, owners, and retirement rules. The artifact should survive both scientific and operational review.

DECISION STANDARD

Reject or redesign any idea whose mechanism, information set, search history, independent evidence, implementation economics, or failure response cannot be stated clearly and reproduced from durable artifacts.

REFERENCE SHEET

Notation and signal contract

BEGINNER

i, t	**instrument or sample and decision time**	stable sample key plus causal cutoff
x_i,t	**raw measured feature**	value, units, availability, coverage and source
s_i,t	**transformed signal score**	sign, scale, clipping, peer group and fitted state
r_i,t->t+h	**future target outcome**	label start, end, price convention and overlap
IC_t	**cross-sectional forecast correlation**	estimator, eligible count, weighting and horizon
w_i,t	**portfolio weight or intended exposure**	constraints, normalization and rebalance clock
c_i,t	**estimated implementation cost**	spread, fees, impact, borrow, delay and uncertainty
theta	**fitted model or transform state**	training window, fold, code and immutable artifact
F_t	**information set available at time t**	source vintage, arrival frontier and entitlement
H_k	**registered hypothesis or branch**	parent idea, search family and primary endpoint

MINIMUM SIGNAL MANIFEST

One score is incomplete without sample identity, decision clock, feature lineage, transform state, sign convention, units, target, eligible population, cost assumptions, uncertainty, and health state. Store these fields beside every result and live decision.

PROCESS FLOW

Reference signal research workflow

BEGINNER

1. **Observe and source**

   Capture the market behavior, source, context, nearest existing idea, and initial counterargument.

2. **Frame the decision**

   Declare universe, clock, action, horizon, benchmark, costs, constraints, and useful effect size.

3. **Specify mechanism**

   Map actors, incentives, causal alternatives, state dependence, decay, capacity, and falsifiers.

4. **Lock hypothesis**

   Predeclare sign, measurement, primary endpoint, split policy, search family, and stop rule.

5. **Build causal sample**

   Pin data vintages, point-in-time eligibility, availability, labels, fitted transforms, and lineage.

6. **Diagnose the signal**

   Inspect support, IC, ordered portfolios, incremental value, decay, costs, and parameter surfaces.

7. **Challenge evidence**

   Apply temporal validation, negative controls, concentration, replication, and adversarial review.

8. **Promote and monitor**

   Grade evidence, prove parity, stage capital, enforce limits, monitor mechanism, and retire safely.

STAGE RULE

A failed stage sends the idea backward for redesign or into the rejected-knowledge archive. It never earns a bypass because later metrics look attractive; evidence strength and control maturity determine the next permitted state.

RESEARCH FRAMING

Decision-first research questions

BEGINNER

ONE-SENTENCE MEANING

A research question is useful only when it names the decision, information cutoff, action, horizon, comparison, and success criterion that the evidence must change. It turns a vague market observation into a falsifiable investigation with a defined consumer.

MEMORY JOGGER

Begin at the capital decision and work backward. Ask what would be traded, when it could be known, how long the exposure would be held, and which result would cause the idea to be rejected.

WHY IT MATTERS IN QUANT RESEARCH

Without a decision contract, researchers can optimize an attractive statistic that has no executable use. A well-framed question constrains labels, features, validation splits, costs, and the eventual portfolio role before coding begins.

FORMULA INTUITION

Decision value can be written as expected net utility of action minus expected net utility of the benchmark action. Predictive accuracy matters only through payoff, risk, cost, and opportunity cost under the stated decision.

SIMPLE MARKET EXAMPLE

Instead of asking whether earnings revisions predict returns, ask whether revisions available by 09:15 can rank liquid US equities for a five-day market-neutral portfolio after conservative open execution costs.

PYTHON / SYSTEM IMPLEMENTATION

Store the question as structured metadata: universe, decision clock, action, forecast target, holding rule, benchmark, constraints, cost model, acceptance thresholds, and owner. Make every experiment inherit this contract.

CONFUSIONS TO AVOID

A broad topic is not a research question, and a backtest objective is not an economic mechanism. Avoid changing the horizon or universe after seeing results without starting a new registered test family.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The question is the root node of the research graph. Data snapshots, features, labels, experiments, validation reports, and promotion decisions should all link back to its immutable identifier.

RELATED CONCEPTS AND QUICK RECALL

Related: decision rule, research mandate, benchmark, acceptance criterion. Recall task: Rewrite one vague alpha idea as a decision contract with an exact clock, action, horizon, benchmark, and rejection rule.

RESEARCH FRAMING

Economic mechanism and behavioral story

BEGINNER

ONE-SENTENCE MEANING

An economic mechanism explains how information, incentives, constraints, or institutions could create a repeatable payoff and why competition has not removed it. It produces testable intermediate predictions, not a story added after success.

MEMORY JOGGER

Name the actor, the pressure, the transmission path, the expected price response, and the reason the effect persists. Then list observations that would contradict each link.

WHY IT MATTERS IN QUANT RESEARCH

Mechanisms guide feature choice, timing, regimes, capacity assumptions, and decay. They also distinguish a portable market effect from an accidental pattern tied to one vendor field or historical sample.

FORMULA INTUITION

A mechanism is a chain: catalyst -> constrained or biased behavior -> order flow or risk transfer -> price adjustment -> decay. Each arrow needs an observable implication and a plausible time scale.

SIMPLE MARKET EXAMPLE

Fund managers benchmarked monthly may rebalance gradually after a large signal update. The story predicts persistent flow, stronger effects in less liquid names, decay near the rebalance horizon, and higher implementation cost where the effect is strongest.

PYTHON / SYSTEM IMPLEMENTATION

Create a mechanism record with actors, incentives, constraints, causal links, predicted cross-sectional signs, timing, state dependence, capacity limits, and disconfirming tests. Review it before exposing outcome metrics.

CONFUSIONS TO AVOID

A plausible story is not evidence, and every price pattern can be rationalized after the fact. Reject stories that do not produce measurements distinct from the final return outcome.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Mechanism metadata connects idea intake to data requirements, subgroup tests, cost scenarios, monitoring signals, and retirement rules. It remains visible in the strategy registry after deployment.

RELATED CONCEPTS AND QUICK RECALL

Related: risk premium, behavioral bias, institutional constraint, causal mechanism. Recall task: For one proposed effect, identify the actor, constraint, transmission channel, decay, and two observations that would falsify the story.

The hypothesis contract

BEGINNER

ONE-SENTENCE MEANING

A hypothesis contract precommits the expected direction, measurement, sample, horizon, evaluation method, and failure conditions before the researcher observes the decisive result. It makes exploration auditable and separates predictions from later explanations.

MEMORY JOGGER

Write what should happen, where, when, compared with what, by how much, and what would make you stop. Ambiguous hypotheses are difficult to disprove and easy to overfit.

WHY IT MATTERS IN QUANT RESEARCH

Explicit contracts reduce silent degrees of freedom in lag choice, transforms, filters, and endpoints. They also let another researcher reproduce the intended test without interpreting notebook history.

FORMULA INTUITION

A useful contract states H1 in an estimand form, such as E[r_future | top signal quintile] - E[r_future | bottom quintile] > threshold, with all conditioning information and costs declared in advance.

SIMPLE MARKET EXAMPLE

A contract may predict that post-announcement drift is positive for large standardized surprises, concentrated in low-coverage names, strongest over days two through ten, and absent before publication time.

PYTHON / SYSTEM IMPLEMENTATION

Serialize hypothesis ID, parent idea, rationale, estimator, data vintage, split policy, primary metric, secondary diagnostics, multiplicity family, minimum effect, and stop rule. Lock primary fields before the confirmatory run.

CONFUSIONS TO AVOID

Statistical significance alone is not a complete success rule. Do not rewrite a two-sided test as directional, promote a secondary slice to primary, or extend the horizon after seeing where the curve peaks.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment service validates that runs conform to the contract and marks deviations as exploratory branches. Reviewers compare the original prediction with the complete outcome set.

RELATED CONCEPTS AND QUICK RECALL

Related: pre-registration, estimand, primary endpoint, researcher degrees of freedom. Recall task: Draft a hypothesis with a signed effect, minimum useful magnitude, primary sample, validation period, multiplicity family, and stop condition.

RESEARCH FRAMING

Causal map and competing explanations

BEGINNER

ONE-SENTENCE MEANING

A causal map records pathways that could generate an observed association, including confounders, mediators, selection effects, and alternatives. It guides controls and negative tests without claiming observational research proves causality.

MEMORY JOGGER

Before testing X against future return, ask what causes X, what else affects both X and return, what is selected into the sample, and which variable might merely sit downstream of price.

WHY IT MATTERS IN QUANT RESEARCH

Many apparent signals proxy for size, liquidity, sector, market beta, stale pricing, or data availability. Mapping alternatives forces the research to measure incremental information rather than rediscover common exposures.

FORMULA INTUITION

The target estimand can be viewed as association remaining after a declared adjustment set, not an unrestricted causal effect. Conditioning on a collider or post-treatment variable can create bias rather than remove it.

SIMPLE MARKET EXAMPLE

A news-intensity feature predicts negative returns, but distressed companies receive more coverage and trade less liquidly. Controls for prior drawdown and liquidity may weaken the effect; controlling for the post-news price move may erase part of the mechanism itself.

PYTHON / SYSTEM IMPLEMENTATION

Represent nodes, arrow direction, observation clocks, and testable implications in a lightweight graph. Translate the graph into baseline controls, residualization choices, subgroup tests, negative controls, and fields that must not enter the model.

CONFUSIONS TO AVOID

More controls are not always safer. A variable can be a mediator, collider, or future observation; automated feature selection does not understand those roles.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The causal map sits beside lineage and model specifications. When a new confounder or data process is discovered, forward lineage identifies which experiments and promoted strategies require reevaluation.

RELATED CONCEPTS AND QUICK RECALL

Related: confounding, mediator, collider, proxy exposure. Recall task: Draw a five-node map for a candidate signal and label one confounder, one mediator, one selection path, and one forbidden future variable.

RESEARCH FRAMING

Universe, horizon, and information set

BEGINNER

ONE-SENTENCE MEANING

The research scope defines which instruments were eligible, which information was knowable at each decision, and which forecast and holding horizons the hypothesis addresses. These choices determine the population represented by every reported statistic.

MEMORY JOGGER

Universe membership is data, not a static filter. Freeze eligibility point in time and keep the decision horizon distinct from the label window and the realized holding period.

WHY IT MATTERS IN QUANT RESEARCH

A result can disappear when delisted names, small securities, overnight information, or realistic publication lags are included. Scope discipline prevents the sample from drifting toward assets and periods that make the idea look strongest.

FORMULA INTUITION

For sample i, require eligibility_i at decision time, max availability(inputs_i) <= cutoff_i, and label_start_i >= executable decision_i. Holding rules may overlap labels and therefore require purged validation.

SIMPLE MARKET EXAMPLE

A daily factor tested on today's index members uses future membership and excludes failed firms. Reconstructing each date's tradable universe may lower returns but yields the population the strategy could actually have traded.

PYTHON / SYSTEM IMPLEMENTATION

Build a versioned sample table containing stable instrument ID, eligibility reason, decision timestamp, data frontier, target interval, planned execution interval, and exclusion codes. Reconcile counts by date and market.

CONFUSIONS TO AVOID

Ticker lists, current classifications, and final corporate-action histories are not historical universes. A one-row lag does not solve publication delays, time zones, auctions, or asynchronous assets.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The sample builder becomes a shared service feeding feature research, labels, backtests, and live eligibility. Every run pins its universe and clock policy rather than recomputing a moving latest view.

RELATED CONCEPTS AND QUICK RECALL

Related: point-in-time universe, decision clock, label horizon, survivorship bias. Recall task: Specify the information set and sample row for a signal decided at the close and executed at the next session's open.

RESEARCH FRAMING

Research priors and base rates

BEGINNER

ONE-SENTENCE MEANING

A research prior records how plausible and useful an idea was before new evidence, based on mechanism quality, related evidence, search breadth, implementation burden, and the historical success rate of similar ideas. It moderates excitement about one favorable sample.

MEMORY JOGGER

Extraordinary backtests are common in large searches and rare in live portfolios. Ask how many similar ideas failed, how flexible the search was, and what evidence would have been expected if the effect were real.

WHY IT MATTERS IN QUANT RESEARCH

Base-rate awareness improves resource allocation and guards against treating every p-value equally. A weakly motivated candidate found among thousands needs stronger replication than a narrowly specified mechanism with independent evidence.

FORMULA INTUITION

Bayesian odds are posterior odds = prior odds x likelihood ratio. The formula is a discipline: evidence updates plausibility; it does not erase the probability of false discovery created by the search process.

SIMPLE MARKET EXAMPLE

A striking Sharpe ratio from an automated library of twenty thousand transformations should receive a lower prior than a predeclared microstructure effect predicted by an exchange rule change and observed in independent venues.

PYTHON / SYSTEM IMPLEMENTATION

Score mechanism, novelty, search size, data quality, external evidence, expected costs, crowding, and operational complexity before confirmatory testing. Preserve the score so outcome knowledge cannot rewrite the prior.

CONFUSIONS TO AVOID

A prior is not permission to dismiss data or a precise subjective probability masquerading as fact. Avoid double-counting evidence that already informed both the story and the backtest specification.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The idea registry uses priors to order expensive investigations, set replication burden, and calibrate promotion thresholds. Portfolio allocation still depends on validated forecast and risk evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: Bayesian updating, false discovery, evidence strength, research portfolio. Recall task: Compare the prior evidence burden for a narrowly predicted venue effect and a best-of-thousands technical pattern.

Structured idea sourcing

BEGINNER

ONE-SENTENCE MEANING

Structured sourcing turns market, institutional, academic, operational, and portfolio observations into traceable candidate mechanisms. It records why an idea entered the queue instead of leaving a backlog of unexplained indicators.

MEMORY JOGGER

Collect anomalies with context: who observed it, under which market state, which actor may cause it, what data would measure it, and how the idea differs from existing exposure.

WHY IT MATTERS IN QUANT RESEARCH

Diverse sources reduce dependence on one discovery style, while structured intake makes duplication, conflicts, and missing evidence visible. It also preserves failed ideas as useful negative knowledge.

FORMULA INTUITION

Idea priority can combine expected decision value, mechanism credibility, data readiness, independence from the current book, estimated capacity, implementation cost, and evidence burden. The score ranks work; it does not validate alpha.

SIMPLE MARKET EXAMPLE

Execution logs show repeated adverse fills after index-close imbalances. The intake record links the symptom to auction mechanics, expected sign by imbalance direction, required order-book fields, and a comparison against ordinary closing sessions.

PYTHON / SYSTEM IMPLEMENTATION

Use an idea template with source, observation date, mechanism, target decision, novel contribution, required data, closest existing idea, initial counterarguments, ethical or license constraints, and triage disposition.

CONFUSIONS TO AVOID

Do not copy a published signal without translating its historical data, market structure, and cost assumptions. Avoid rewarding idea quantity when the registry contains near-duplicate transforms of the same exposure.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The intake queue feeds a research council or owner who deduplicates, scores, rejects, parks, or converts candidates into hypothesis contracts. All downstream artifacts retain the source record.

RELATED CONCEPTS AND QUICK RECALL

Related: idea backlog, literature replication, portfolio diagnostics, research triage. Recall task: Turn a trader observation into a complete intake record and identify the nearest existing exposure it may duplicate.

Behavioral and attention signals

BEGINNER

ONE-SENTENCE MEANING

Behavioral signals seek predictable responses to underreaction, overreaction, anchoring, limited attention, delegated incentives, or crowded decision rules. Their credibility depends on observing the behavior or a distinctive implication, not merely naming a bias.

MEMORY JOGGER

A behavioral label should predict who reacts slowly or excessively, when attention is scarce, and how the mispricing closes. Test those cross-sectional and timing implications.

WHY IT MATTERS IN QUANT RESEARCH

Behavioral effects can persist because information processing and incentives are costly, but they can reverse as technology, participation, or regulation changes. They often interact with liquidity and event salience.

FORMULA INTUITION

A useful design compares future payoff conditional on both signal strength and a mechanism proxy such as coverage, distraction, ownership, or complexity. The interaction should be specified before inspecting every possible subgroup.

SIMPLE MARKET EXAMPLE

A complex filing released during a crowded announcement day may be processed more slowly. The mechanism predicts stronger drift where analyst coverage is low and weaker effects for simple, highly followed firms.

PYTHON / SYSTEM IMPLEMENTATION

Construct point-in-time attention and complexity proxies, retain their coverage and source clocks, and compare incremental forecasts after size, sector, prior return, and liquidity controls. Test decay around the relevant event.

CONFUSIONS TO AVOID

Post-hoc references to underreaction do not validate a signal. Attention proxies can encode firm size or data availability, and social data may be manipulated, revised, or licensed inconsistently.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Behavioral candidates enter the event and feature layers, while mechanism-specific monitors track participation, salience, and decay. Retirement review asks whether the behavioral bottleneck still exists.

RELATED CONCEPTS AND QUICK RECALL

Related: limited attention, underreaction, overreaction, salience. Recall task: Design one mechanism-specific test that would distinguish limited attention from a generic risk or liquidity premium.

Structural and institutional signals

BEGINNER

ONE-SENTENCE MEANING

Structural signals arise from rules, mandates, funding constraints, index mechanics, dealer inventory, hedging needs, market design, or operational schedules that create predictable demand or risk transfer. The rule and affected actor should be identifiable.

MEMORY JOGGER

Look for forced or price-insensitive flows: who must trade, under what trigger, on which clock, with what flexibility, and against which liquidity providers.

WHY IT MATTERS IN QUANT RESEARCH

Institutional mechanisms can be more testable than broad behavioral stories, but they are exposed to rule changes, anticipation, crowding, and execution competition. Capacity is often concentrated near the event.

FORMULA INTUITION

Expected net effect depends on forced flow relative to available liquidity, anticipation by other traders, and the inventory cost of absorbing the imbalance. Scaling by ordinary market activity often matters more than the raw flow estimate.

SIMPLE MARKET EXAMPLE

An index rebalance creates a known demand schedule for additions and deletions. A strategy might provide liquidity before the close, but profitability depends on announcement timing, passive assets, auction depth, and crowding.

PYTHON / SYSTEM IMPLEMENTATION

Version rule documents and calendars, estimate affected holdings or hedges, map the full event timeline, and replay order-book conditions. Build scenarios for early anticipation, partial completion, and rule amendment.

CONFUSIONS TO AVOID

A predictable flow is not automatically a profitable trade because prices may adjust before execution. Historical event timestamps are often backfilled to their effective date rather than the date they became public.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The event service supplies rule versions and deadlines; the research layer estimates impact; execution simulates competition; monitoring watches participation and post-event decay for structural change.

RELATED CONCEPTS AND QUICK RECALL

Related: forced flow, index rebalance, dealer inventory, market design. Recall task: Map the announcement, anticipation, execution, and reversal stages of one institutional flow and name the executable decision point.

Risk premia versus mispricing

BEGINNER

ONE-SENTENCE MEANING

A signal payoff may compensate for bearing undesirable risk, capture a temporary mispricing, or mix both. Distinguishing the interpretations changes leverage, hedging, stress expectations, and how much decay should be expected after discovery.

MEMORY JOGGER

Ask whether the payoff survives factor controls, appears in bad states, requires scarce capital, and is earned by providing insurance. A high average return can be payment for losses exactly when capital is valuable.

WHY IT MATTERS IN QUANT RESEARCH

Risk-premium strategies may remain viable but demand drawdown tolerance and capital reserves; mispricing strategies may decay through competition and depend on correction catalysts. Portfolio construction should not treat them as interchangeable alphas.

FORMULA INTUITION

Decompose signal return as intended common-risk exposure plus residual payoff plus implementation effects. The decomposition is model-dependent, so report uncertainty and alternative risk models.

SIMPLE MARKET EXAMPLE

A short-volatility signal earns steady carry and crashes during stress. Calling the carry pure alpha ignores its insurance-writing role; hedging the tail may reveal whether any residual forecast remains.

PYTHON / SYSTEM IMPLEMENTATION

Run factor and scenario attribution, conditional performance in funding and volatility states, drawdown-shape analysis, and hedged variants. Record the economic service supplied to counterparties and the capital required in stress.

CONFUSIONS TO AVOID

Factor neutrality under one linear model does not prove mispricing, and a crash exposure does not make a strategy invalid. Avoid removing the very mechanism the strategy is designed to monetize without noting the changed question.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The classification informs risk budgets, benchmark choice, scenario library, capacity assumptions, and investor communication. Registry metadata should allow a strategy to have mixed rationale with stated confidence.

RELATED CONCEPTS AND QUICK RECALL

Related: risk premium, anomaly, factor exposure, insurance payoff. Recall task: Explain whether a steady carry strategy with rare severe losses is likely alpha, risk compensation, or a mixture, and name tests that separate them.

Time-series and cross-sectional formulations

BEGINNER

ONE-SENTENCE MEANING

A time-series signal forecasts an instrument relative to its own history, while a cross-sectional signal ranks instruments against contemporaneous peers. The formulations answer different questions and impose different neutrality, scaling, and data requirements.

MEMORY JOGGER

Time-series asks whether this asset should be long or short now; cross-sectional asks which assets should be owned relative to others now. The same raw feature can support either design.

WHY IT MATTERS IN QUANT RESEARCH

The distinction determines label structure, standardization, portfolio constraints, and exposure to common market moves. Combining their metrics without a declared formulation can hide where returns originate.

FORMULA INTUITION

Time-series score may be $f_{i,t}$ divided by its trailing scale; cross-sectional score may be $\mathrm{rank}_t(f_{i,t})$ within an eligible peer group. The first uses historical calibration, the second depends on the current population.

SIMPLE MARKET EXAMPLE

A twelve-month return can trigger a long position in every asset during a broad uptrend, or it can rank winners against losers in a dollar-neutral portfolio. These strategies have materially different beta and crisis behavior.

PYTHON / SYSTEM IMPLEMENTATION

Implement separate sample builders and transformations, then evaluate market exposure, breadth, turnover, missingness, and capacity. Preserve raw values so the formulations can be compared on identical information.

CONFUSIONS TO AVOID

Cross-sectional ranking does not eliminate sector or beta exposure automatically, and time-series normalization can drift when volatility regimes change. A rank also discards magnitude that may matter for sizing.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The strategy specification declares formulation before portfolio construction. Shared feature artifacts may feed both branches, but validation and governance treat them as distinct hypotheses.

RELATED CONCEPTS AND QUICK RECALL

Related: relative ranking, directional forecast, peer group, market neutrality. Recall task: Convert one raw momentum feature into both formulations and state the exposure introduced by each.

Signal direction, scale, and monotonicity

BEGINNER

ONE-SENTENCE MEANING

Signal design converts a measured feature into a signed forecast or desired ordering with declared direction, scale, clipping, and expected response shape. Those choices should reflect the mechanism and be fitted only on training evidence.

MEMORY JOGGER

A feature is not yet a tradable signal. State whether larger values imply larger returns, whether the relation saturates or reverses, and which unit one score point represents.

WHY IT MATTERS IN QUANT RESEARCH

Direction errors, unstable scales, and arbitrary nonlinear transforms can dominate performance. A monotonic response is easier to validate and monitor than a complex shape discovered through flexible tuning.

FORMULA INTUITION

A standardized signal can be $s=(x-\text{center})/\text{scale}$ followed by a predeclared mapping $q(s)$. Forecast calibration estimates $E[r_{future} \mid s]$ in training data, with uncertainty and support reported across the range.

SIMPLE MARKET EXAMPLE

If a valuation spread is expected to mean revert, a larger positive deviation may imply a short forecast. Extreme deviations may be capped when borrow, distress, or measurement error breaks linearity.

PYTHON / SYSTEM IMPLEMENTATION

Plot binned outcomes with counts and intervals, fit center and scale within folds, compare raw, rank, clipped, and monotone mappings, and persist parameters. Monitor score distribution and realized calibration live.

CONFUSIONS TO AVOID

Choosing the sign after viewing results consumes a test, and forcing monotonicity can hide a genuine threshold or reversal. Conversely, a wavy fitted curve with sparse tails is usually sampling noise.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The signal artifact binds feature lineage, transform state, sign convention, calibration, coverage, and health metrics. Portfolio sizing consumes the forecast in consistent return and risk units.

RELATED CONCEPTS AND QUICK RECALL

Related: forecast calibration, rank transform, winsorization, monotone mapping. Recall task: Given a U-shaped response, decide whether theory supports a magnitude signal or whether the result should be treated as an unstable discovery.

Exploratory analysis without silent selection

BEGINNER

ONE-SENTENCE MEANING

Exploratory analysis studies data quality, support, timing, distributions, and candidate relationships while preserving a record of every material choice. It is legitimate discovery work, but its outcomes require separate confirmation.

MEMORY JOGGER

Explore broadly, log completely, and confirm narrowly. Treat each inspected transform, lag, subgroup, and chart as information consumed from the development sample.

WHY IT MATTERS IN QUANT RESEARCH

Visual inspection can reveal defects and mechanisms that a summary metric misses, yet repeated inspection also tunes the researcher. Honest labeling protects the final holdout and makes search breadth measurable.

FORMULA INTUITION

The effective trial count includes formal models plus human choices influenced by outcomes. No single formula captures it perfectly, so store the full path and use stronger out-of-sample evidence when exploration was flexible.

SIMPLE MARKET EXAMPLE

A heatmap across horizons shows an effect only at seven days. That observation can generate a new seven-day hypothesis, but the same heatmap cannot serve as confirmatory evidence for the newly selected peak.

PYTHON / SYSTEM IMPLEMENTATION

Capture notebook or pipeline version, viewed metrics, filters, plots, transforms, and decision notes in an experiment log. Reserve a sealed period or independent market for the final specification.

CONFUSIONS TO AVOID

Exploration is not misconduct, and pre-registration is not a substitute for understanding data. The error is reporting a selected exploratory result as if its specification had been fixed beforehand.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experimentation platform labels runs exploratory, confirmatory, replication, or monitoring. Holdout permissions and result visibility can be controlled by phase.

RELATED CONCEPTS AND QUICK RECALL

Related: data snooping, holdout, garden of forking paths, exploratory analysis. Recall task: Explain how a horizon heatmap can generate a valid new hypothesis without validating the selected horizon itself.

Information coefficient and forecast correlation

BEGINNER

ONE-SENTENCE MEANING

The information coefficient measures association between a signal and a future outcome, commonly by date across instruments or through time for one instrument. Its interpretation requires the estimator, horizon, sample size, overlap, and weighting scheme.

MEMORY JOGGER

IC asks whether higher scores tend to align with higher future outcomes, not whether the strategy is profitable after sizing and costs. Inspect its distribution, not only its average.

WHY IT MATTERS IN QUANT RESEARCH

IC is a compact early diagnostic for direction, stability, and breadth, but serial dependence and overlapping labels reduce effective observations. Rank IC is robust to monotone scaling but still sensitive to universe construction.

FORMULA INTUITION

For date t, $IC_t = corr(s_{i,t}, r_{i,t+h})$ across eligible i. Report mean, standard deviation, hit rate, autocorrelation, Newey-West or block uncertainty, and weighted alternatives when relevant.

SIMPLE MARKET EXAMPLE

A signal has average rank IC 0.03 across 800 dates, but most value comes from a small-cap subgroup during one regime. Aggregate significance can hide concentration and poor capacity.

PYTHON / SYSTEM IMPLEMENTATION

Compute daily cross-sectional IC with point-in-time eligibility, preserve per-date counts, and summarize by era, sector, liquidity, and market state. Use block bootstrap or appropriate dependence-aware intervals.

CONFUSIONS TO AVOID

A high IC does not guarantee a high portfolio Sharpe because turnover, covariance, constraints, tails, and costs intervene. Pooling all rows can overweight dates with more instruments and create misleading precision.

WHERE THIS FITS IN THE RESEARCH SYSTEM

IC diagnostics sit between signal construction and portfolio tests. The registry stores exact target, correlation type, weighting, sample policy, and uncertainty method.

RELATED CONCEPTS AND QUICK RECALL

Related: rank IC, Pearson correlation, effective sample size, breadth. Recall task: List the fields required to interpret a reported IC of 0.04 and explain why a pooled row-level correlation answers a different question.

Quantile portfolios and monotonicity

BEGINNER

ONE-SENTENCE MEANING

Quantile portfolios group observations by signal rank and compare future outcomes across the ordered groups. They reveal monotonicity, tail concentration, asymmetry, and economic magnitude without assuming a linear response.

MEMORY JOGGER

Do not focus only on top minus bottom. Read the entire ordered profile, counts, uncertainty, turnover, and exposure of every bucket.

WHY IT MATTERS IN QUANT RESEARCH

A smooth gradient supports a scalable ranking interpretation, while a single extreme bucket may indicate threshold economics, outliers, distress, or selection. Bucket profiles also reveal whether neutralization changes the signal.

FORMULA INTUITION

At each rebalance, assign eligible observations using training-defined or cross-sectional breakpoints, compute weighted bucket returns, and form declared spreads. Inference must respect time dependence and overlapping holdings.

SIMPLE MARKET EXAMPLE

A five-bin result of -2, -1, 0, 1, and 2 basis points per day supports ordering; a result of 0, 0, 0, 0, and 6 may be useful but demands tail coverage, capacity, and outlier tests.

PYTHON / SYSTEM IMPLEMENTATION

Publish bucket means, medians, intervals, counts, turnover, factor exposures, costs, and cumulative returns. Repeat with fixed breakpoints and value or liquidity weights to test dependence on portfolio construction.

CONFUSIONS TO AVOID

Equal-count buckets can contain very different score distances, and sorting after excluding hard-to-trade names changes the estimand. Top-bottom significance alone can hide a nonmonotonic middle.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Quantile diagnostics inform mapping and sizing but do not replace an executable backtest. The approved signal contract records the chosen breakpoints and portfolio interpretation.

RELATED CONCEPTS AND QUICK RECALL

Related: portfolio sort, breakpoint, monotonicity, long-short spread. Recall task: Interpret two five-bucket profiles: one smooth and one driven only by the extreme tail, including the next validation step for each.

Incremental value and baseline controls

BEGINNER

ONE-SENTENCE MEANING

Incremental-value testing asks whether a candidate adds forecast information or decision improvement beyond a strong, predeclared baseline containing known exposures and existing signals. It prevents renaming old risk as a new idea.

MEMORY JOGGER

Beat the right benchmark. Compare against the production stack or a credible simple model, not against zero information.

WHY IT MATTERS IN QUANT RESEARCH

Many candidates correlate with momentum, size, volatility, sector, liquidity, or current portfolio positions. A modest incremental improvement can matter more than an impressive standalone result that duplicates crowded exposure.

FORMULA INTUITION

Evaluate nested models or residualized signals using out-of-sample delta loss, delta IC, delta utility, and conditional contribution. Uncertainty should cover the paired difference because both models use the same observations.

SIMPLE MARKET EXAMPLE

A text signal has IC 0.06 alone but 0.01 after controlling for revisions and price momentum. Its independent value may still exist in low-coverage firms, but the broad headline overstates novelty.

PYTHON / SYSTEM IMPLEMENTATION

Freeze baseline version, fit all controls inside folds, compare paired predictions, inspect coefficients and contribution by slice, and measure portfolio overlap. Include a simple baseline that exposes unnecessary complexity.

CONFUSIONS TO AVOID

Residualization can remove a mechanism if the control lies downstream, and a stronger baseline can itself be overfit. Do not change the benchmark after seeing which comparison the candidate wins.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model registry links candidate and baseline artifacts. Promotion reports show standalone and incremental evidence, overlap with active strategies, and marginal capital impact.

RELATED CONCEPTS AND QUICK RECALL

Related: nested model, residualization, ablation, forecast combination. Recall task: Design a paired out-of-sample comparison between a candidate and the current signal stack, including one portfolio-level overlap metric.

Turnover, decay, and implementability

BEGINNER

ONE-SENTENCE MEANING

Implementability diagnostics connect forecast decay to trading speed, turnover, spread, impact, borrow, latency, and position constraints. They ask whether the market opportunity remains after an executable path from signal to position.

MEMORY JOGGER

A forecast has a half-life; a trade has a clock and a cost curve. Profit requires enough signal to survive delay, rebalance, and unwind under realistic capacity.

WHY IT MATTERS IN QUANT RESEARCH

Short-lived signals can look excellent at frictionless timestamps and fail after one bar of delay. Slow signals may tolerate patient execution but tie up capital and overlap with existing positions.

FORMULA INTUITION

Net forecast at delay d is expected gross payoff remaining after d minus spread, fees, impact, borrow, and risk cost. Turnover is traded absolute weight relative to capital under a declared rebalance convention.

SIMPLE MARKET EXAMPLE

A signal earns 8 basis points immediately but decays to 3 after thirty minutes; conservative round-trip cost is 4. The opportunity is negative for that execution path despite a strong contemporaneous forecast metric.

PYTHON / SYSTEM IMPLEMENTATION

Estimate decay by event time, simulate multiple delays and participation rates, join point-in-time quotes and borrow, and report gross-to-net waterfalls. Stress spreads, impact, and fill probability in difficult regimes.

CONFUSIONS TO AVOID

Average spread is not total cost, volume is not guaranteed capacity, and observed close prices are not always executable. Turnover computed after netting can hide gross operational burden.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Early research uses a conservative cost screen; backtesting adds order logic; execution research calibrates fills; portfolio construction controls turnover and capacity using the same signal clock.

RELATED CONCEPTS AND QUICK RECALL

Related: signal decay, half-life, transaction cost, market impact. Recall task: Given an eight-basis-point instantaneous forecast and a four-basis-point cost, determine which decay measurements are needed before calling it viable.

Parameter surfaces and sensitivity

BEGINNER

ONE-SENTENCE MEANING

A parameter surface shows how results change across economically meaningful choices such as lookback, holding horizon, threshold, neutralization, and rebalance frequency. Robust ideas usually occupy a coherent region rather than one isolated optimum.

MEMORY JOGGER

Prefer plateaus to peaks. The neighborhood around the selected specification reveals whether performance reflects a stable mechanism or a lucky coordinate.

WHY IT MATTERS IN QUANT RESEARCH

Sensitivity analysis exposes hidden tuning, supports operational tolerances, and identifies where implementation changes might invalidate the edge. It also makes the total search family visible for multiplicity control.

FORMULA INTUITION

For parameter vector theta, inspect out-of-sample metric M(theta), local gradients, rank stability, and worst-neighbor performance. Selection must still occur inside training or nested validation.

SIMPLE MARKET EXAMPLE

A moving-average strategy works only at lookbacks 37 and 113 days, while nearby choices fail. Without a mechanism explaining those exact values, the twin peaks are weak evidence.

PYTHON / SYSTEM IMPLEMENTATION

Define a bounded grid from mechanism and operational constraints, compute every cell with identical data and splits, save the full surface, and select a simple representative within a stable region. Recheck on later data.

CONFUSIONS TO AVOID

A smooth surface can arise because neighboring tests share data, so it does not prove truth. Searching ever-larger grids and showing only the final neighborhood hides the real trial count.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Surface artifacts guide robust defaults and alert thresholds. The experiment registry records the complete grid, selection rule, and chosen cell before the sealed evaluation.

RELATED CONCEPTS AND QUICK RECALL

Related: hyperparameter tuning, robustness plateau, nested validation, specification curve. Recall task: Choose between a slightly lower broad plateau and the highest isolated cell, then explain how nested validation preserves honest evidence.

Multiple testing and false discovery

BEGINNER

ONE-SENTENCE MEANING

Multiple testing recognizes that large searches produce impressive results by chance and adjusts the evidence standard for the number and dependence of attempted ideas. The relevant family includes hidden transforms, lags, filters, and human-selected views.

MEMORY JOGGER

The winner of many noisy trials is biased upward. Record every trial and reserve independent evidence rather than trusting the best in-sample statistic.

WHY IT MATTERS IN QUANT RESEARCH

Quant research naturally searches broad spaces, making false discovery a primary operational risk. Controlling it protects research capacity, model complexity, and capital from statistical accidents.

FORMULA INTUITION

For m tests, Bonferroni uses alpha/m for family-wise error; Benjamini-Hochberg compares ordered p-values with q*i/m for false-discovery control. Dependence and sequential selection affect which method is appropriate.

SIMPLE MARKET EXAMPLE

At a nominal 5 percent threshold, one thousand null candidates produce about fifty false positives on average. Selecting the top five and reporting only those conceals the original evidence burden.

PYTHON / SYSTEM IMPLEMENTATION

Assign search-family IDs, log all specifications, use dependence-aware corrections where possible, apply deflated or resampled performance benchmarks, and require a sealed temporal or cross-market replication.

CONFUSIONS TO AVOID

Correction does not rescue poor timing, unstable data, or an implausible mechanism. Bonferroni may be conservative, but that is not a rationale for ignoring multiplicity or informal analyst tuning.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry computes family-level evidence and prevents isolated runs from being promoted without their search context. Governance reviews both statistical correction and independent replication.

RELATED CONCEPTS AND QUICK RECALL

Related: family-wise error, false discovery rate, selection bias, replication. Recall task: Estimate expected false positives from a thousand null tests and describe how you would structure an independent confirmation.

Temporal validation and walk-forward evidence

BEGINNER

ONE-SENTENCE MEANING

Temporal validation trains and selects using earlier information, then evaluates on later periods with folds that mimic how the strategy would have been refreshed. Walk-forward evidence reveals stability across vintages rather than one favorable train-test split.

MEMORY JOGGER

Time does not shuffle. Respect label overlap, feature windows, data revisions, and retraining cadence so validation reproduces the information boundary of live research.

WHY IT MATTERS IN QUANT RESEARCH

Markets change, observations are dependent, and overlapping labels leak across nearby samples. A sequence of honest folds shows whether the effect survived multiple decision eras and model updates.

FORMULA INTUITION

For each fold k, fit transformations and models on training times, optionally tune on a later validation block, purge overlap around boundaries, and score only the forward test block. Aggregate fold-level metrics with their heterogeneity.

SIMPLE MARKET EXAMPLE

A five-day label using observations through Friday cannot be in the training set when the test starts Monday if the outcome window crosses the boundary. Purging and embargo remove that shared future information.

PYTHON / SYSTEM IMPLEMENTATION

Implement a fold generator driven by timestamps and label intervals, bind fitted artifacts to each fold, and preserve per-fold predictions before aggregation. Compare expanding and rolling training windows when the mechanism suggests adaptation.

CONFUSIONS TO AVOID

Walk-forward testing does not guarantee realism if the data vintage is final or hyperparameters are chosen from all folds. Treating folds as independent can also overstate precision.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Temporal validation produces the primary evidence package. Approved retraining schedules, feature states, and model artifacts should mirror the same fit and score boundaries in production.

RELATED CONCEPTS AND QUICK RECALL

Related: walk-forward validation, purging, embargo, rolling window. Recall task: Design three chronological folds for a five-day target and state where purging, tuning, and final model fitting occur.

Regime, cohort, and concentration analysis

BEGINNER

ONE-SENTENCE MEANING

Concentration analysis decomposes results across time, instruments, sectors, liquidity, market states, event types, and trade clusters to determine whether evidence is broad or dependent on a narrow episode. Slices should be motivated and denominators preserved.

MEMORY JOGGER

Find who paid, when they paid, and whether the strategy could survive without its best period or smallest cohort. Broad average performance can conceal one unrepeatable source.

WHY IT MATTERS IN QUANT RESEARCH

A signal may be economically valid only in a declared state, or it may merely overfit a crisis. Concentration diagnostics guide conditional use, capital limits, and the burden of replication.

FORMULA INTUITION

Report contribution shares, leave-one-period-out results, effective number of contributors, worst cohort, and interaction with predeclared states. Uncertainty grows quickly as slices become small.

SIMPLE MARKET EXAMPLE

Seventy percent of P&L comes from one sector during a two-month rebound. The result may reflect a hidden beta or event exposure rather than a persistent stock-selection effect.

PYTHON / SYSTEM IMPLEMENTATION

Build a standard slice cube from point-in-time attributes, then add only mechanism-specific cohorts. Preserve counts, capital, turnover, exposure, and cost for every slice and compare with a simple benchmark.

CONFUSIONS TO AVOID

Searching many slices for the one that works repeats the multiple-testing problem. A regime label created with future data can leak, and removing the worst episode after inspection biases results upward.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evidence report feeds portfolio limits, gating rules, and monitors. A promoted conditional strategy declares the state variable, its live availability, and behavior outside the active state.

RELATED CONCEPTS AND QUICK RECALL

Related: regime analysis, contribution concentration, subgroup, effective breadth. Recall task: Diagnose a strategy whose profit is dominated by one sector and one quarter without automatically deleting that evidence.

Negative controls and placebo tests

BEGINNER

ONE-SENTENCE MEANING

Negative controls and placebos apply a similar test where the mechanism predicts no effect, helping reveal leakage, common confounding, bad timing, or a pipeline that generates significance too easily. They are designed from the causal story.

MEMORY JOGGER

A credible mechanism should say not only where the signal works, but where it should not work. Test the absence prediction with equal rigor.

WHY IT MATTERS IN QUANT RESEARCH

Placebos can catch defects that ordinary robustness checks miss, especially when every candidate benefits from the same timestamp error or universe construction. They strengthen interpretation when they pass and trigger investigation when they do not.

FORMULA INTUITION

Examples include future-shift placebos, pre-event outcomes, unaffected instruments, randomized assignments preserving dependence, and mechanism-irrelevant events. Null distributions must respect serial and cross-sectional structure.

SIMPLE MARKET EXAMPLE

A signal based on earnings publication should not predict returns before the release. If pre-release performance matches post-release performance, the event timestamp or sample selection is probably contaminated.

PYTHON / SYSTEM IMPLEMENTATION

Register expected-null tests with the hypothesis, use block or cluster randomization when appropriate, and retain exact placebo specifications. Compare pipeline behavior, not only p-values, between real and control samples.

CONFUSIONS TO AVOID

A failed placebo does not identify the cause by itself, and an underpowered placebo is weak assurance. Randomly shuffling time-series rows destroys dependence and can create an unrealistically easy null.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Placebo results are mandatory evidence objects linked to the primary test. Pipeline certification can block the whole feature family when a control reveals systemic leakage.

RELATED CONCEPTS AND QUICK RECALL

Related: negative control, permutation test, pre-trend, falsification. Recall task: Design one time placebo, one cross-sectional negative control, and one dependence-preserving randomized test for an event signal.

Independent replication and portability

BEGINNER

ONE-SENTENCE MEANING

Independent replication reruns a fixed specification on a new period, market, venue, dataset, or implementation not used in discovery. Portability analysis asks which parts of the mechanism survive and which depend on local structure.

MEMORY JOGGER

Freeze the recipe before crossing the boundary. Replication is strongest when both the data and the person or code path are meaningfully independent.

WHY IT MATTERS IN QUANT RESEARCH

A successful replication reduces the probability of sample-specific noise and exposes undocumented assumptions. A failed replication is informative when differences in actors, rules, or measurement are analyzed rather than hidden.

FORMULA INTUITION

Compare signed effect, confidence interval, calibration, turnover, cost, and mechanism-specific intermediate measures. Pooling results is appropriate only after testing heterogeneity and data comparability.

SIMPLE MARKET EXAMPLE

A closing-auction effect found in one venue is tested in another with different imbalance publication rules. A weaker effect may support the mechanism if it scales with the transparency and passive-flow differences predicted in advance.

PYTHON / SYSTEM IMPLEMENTATION

Package immutable code, schemas, transform state, and expected outputs; have the replication environment map only declared local fields. Record deviations before result access and reconcile sample coverage.

CONFUSIONS TO AVOID

Applying the same code to a highly correlated period is not fully independent, and changing parameters to fit the new market converts replication into fresh exploration. Failure does not automatically refute a mechanism when market structure differs.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Replication artifacts feed evidence grading and deployment scope. The registry marks which markets and data providers are supported rather than generalizing one result globally.

RELATED CONCEPTS AND QUICK RECALL

Related: external validity, reproducibility, heterogeneity, transportability. Recall task: Specify a replication boundary for a venue-specific signal and list which differences should be mapped before opening results.

Adversarial review and red-team testing

BEGINNER

ONE-SENTENCE MEANING

Adversarial review assigns a researcher or reviewer to find the strongest alternative explanation, data defect, leakage path, cost omission, instability, or operational failure that could invalidate the idea. The goal is decision quality, not defense of sunk work.

MEMORY JOGGER

Ask what would make this backtest easiest to fake accidentally. Attack clocks, universes, labels, search history, costs, concentration, code paths, and the mechanism's most fragile link.

WHY IT MATTERS IN QUANT RESEARCH

Independent challenge counters ownership bias and uncovers assumptions normalized by the original author. It is particularly valuable for high-complexity, high-capital, or unusually strong results.

FORMULA INTUITION

A red-team plan can score severity times plausibility times detectability for each failure mode, then prioritize tests that most reduce decision uncertainty. Scores organize work rather than estimate exact risk.

SIMPLE MARKET EXAMPLE

A reviewer notices that a news feature uses the vendor's corrected publication timestamp and asks for raw arrival logs. The strategy loses most of its edge when replayed with first-seen times, revealing look-ahead contamination.

PYTHON / SYSTEM IMPLEMENTATION

Provide read-only access to the full experiment trail, raw sample evidence, and code. Require written attack hypotheses, reproducible tests, author responses, unresolved risks, and explicit disposition by an independent approver.

CONFUSIONS TO AVOID

Red teaming should not become vague skepticism or an unbounded request for every possible test. The original author should not control which failures are material enough to disclose.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The promotion workflow assigns challenge ownership and tracks every finding to closure, acceptance, scope limitation, or rejection. Serious findings propagate through lineage to related research.

RELATED CONCEPTS AND QUICK RECALL

Related: model risk, independent validation, challenge process, failure mode. Recall task: Draft five prioritized attack tests for an unusually high-Sharpe event strategy and define what evidence would stop promotion.

Economic significance and decision value

BEGINNER

ONE-SENTENCE MEANING

Economic significance measures whether the forecast changes a real decision enough to justify risk, cost, complexity, and scarce capital. It translates statistical evidence into expected net improvement under constraints and uncertainty.

MEMORY JOGGER

A tiny reliable effect may be valuable at scale; a large noisy effect may be unusable. Evaluate net contribution, capacity, diversification, drawdown, and operational burden together.

WHY IT MATTERS IN QUANT RESEARCH

Statistical metrics are intermediate evidence. Portfolio value depends on covariance with current exposures, implementation cost, capital consumption, tail behavior, and whether the forecast improves decisions at the margin.

FORMULA INTUITION

Incremental value can be approximated by change in expected return minus risk penalty, trading cost, financing, and operational cost, evaluated under forecast uncertainty and realistic constraints.

SIMPLE MARKET EXAMPLE

A candidate adds only 15 basis points of annual standalone return but diversifies the book during trend reversals. Its marginal portfolio utility may exceed that of a higher-return signal already correlated with active positions.

PYTHON / SYSTEM IMPLEMENTATION

Run constrained portfolio comparisons with and without the candidate, use identical risk and cost assumptions, perturb forecast strength, and report marginal utility, capital use, stress loss, turnover, and implementation effort.

CONFUSIONS TO AVOID

Multiplying IC by breadth is not a complete business case, and annualizing a short sample can exaggerate precision. Avoid counting diversification benefits from an unstable covariance estimate.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The investment committee or automated allocation service consumes the economic-value report, not just the research metric. Accepted uncertainty determines pilot size and monitoring thresholds.

RELATED CONCEPTS AND QUICK RECALL

Related: marginal utility, capacity, diversification, forecast uncertainty. Recall task: Compare a high-return redundant signal with a lower-return diversifier using marginal portfolio value rather than standalone Sharpe.

Point-in-time provenance and lineage

BEGINNER

ONE-SENTENCE MEANING

Research provenance records the exact source snapshots, availability clocks, transformations, code, parameters, environment, and parents that produced every sample, feature, prediction, and result. It makes historical evidence replayable after data corrections.

MEMORY JOGGER

Never let latest be the only address. A result without pinned evidence and transformation state cannot support a durable strategy decision.

WHY IT MATTERS IN QUANT RESEARCH

Market data are revised, universes change, and pipelines evolve. Provenance distinguishes genuine model drift from changed inputs and lets a correction be traced to affected hypotheses and portfolios.

FORMULA INTUITION

Artifact identity should hash behaviorally relevant parents: data snapshots, schema, clock policy, code, configuration, fitted state, random seed, and environment. Two equal tables from different histories may still need distinct lineage.

SIMPLE MARKET EXAMPLE

A vendor corrects an earnings timestamp. Forward lineage identifies the feature matrices, backtests, review reports, and candidate strategies that used the prior version, allowing targeted reruns instead of broad uncertainty.

PYTHON / SYSTEM IMPLEMENTATION

Publish immutable manifests with row counts, checksums, sample keys, parents, and quality status. Add future-perturbation and replay tests that reconstruct representative dates from raw evidence.

CONFUSIONS TO AVOID

A Git commit alone does not identify data, parameters, external libraries, or fitted transforms. Re-running a query against a mutable table is not reproduction of the original information set.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The artifact registry is the system spine connecting idea, hypothesis, data, feature, model, validation, backtest, approval, and live package. Corrections use lineage to scope impact.

RELATED CONCEPTS AND QUICK RECALL

Related: artifact manifest, data vintage, reproducibility, lineage graph. Recall task: List every parent needed to reproduce one daily signal score exactly six months after a vendor correction.

Experiment registry and search accounting

BEGINNER

ONE-SENTENCE MEANING

An experiment registry stores every material specification, outcome, search-family relationship, decision, and deviation so the organization can reconstruct how a candidate was selected. It is both a scientific record and a control against repeated hidden testing.

MEMORY JOGGER

If a result influenced a choice, record it. Failed and abandoned runs are evidence about the search process, not clutter to delete.

WHY IT MATTERS IN QUANT RESEARCH

Complete histories enable multiplicity correction, prevent duplicated work, reveal analyst tuning, and preserve lessons from rejected mechanisms. They also support handoff when people or tooling change.

FORMULA INTUITION

A run identity includes hypothesis, parents, code, configuration, random state, splits, metrics, artifacts, and status. Search families connect related lags, transforms, models, and manual branches to one evidence budget.

SIMPLE MARKET EXAMPLE

Twenty variations of a reversal feature appear as separate successful notebooks. A registry shows they belong to one family of 240 attempted variants, materially changing the interpretation of the top result.

PYTHON / SYSTEM IMPLEMENTATION

Capture runs automatically from pipelines, allow written notes and dispositions, prevent destructive edits, and expose family-level summaries. Require explicit reasons for cancellation, deviation, promotion, and retirement.

CONFUSIONS TO AVOID

Logging only production candidates creates survivorship bias, while logging every low-level debug action can obscure material decisions. Define what constitutes a trial before the project begins.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry links idea intake, hypothesis locks, compute jobs, review comments, evidence grades, model packages, and live monitoring. Governance queries it directly rather than relying on slide summaries.

RELATED CONCEPTS AND QUICK RECALL

Related: experiment tracking, search family, audit trail, selection history. Recall task: Define the boundary between a debug run and a research trial, then explain how hidden manual chart choices are recorded.

Evidence grading and promotion gates

BEGINNER

ONE-SENTENCE MEANING

Evidence grading summarizes mechanism, data integrity, statistical strength, replication, robustness, costs, capacity, operations, and unresolved risk. A gate requires minimum evidence and named authority before capital exposure increases.

MEMORY JOGGER

Promotion is not a single metric threshold. The evidence package must show why the effect should exist, whether it survived honest tests, and how it will fail safely.

WHY IT MATTERS IN QUANT RESEARCH

Strong backtests can coexist with weak mechanisms, fragile data, or untested operations. Multi-dimensional grading prevents one exceptional statistic from compensating silently for a critical control gap.

FORMULA INTUITION

A grade can combine veto conditions with scored dimensions. Data leakage, unreproducible lineage, prohibited licensing, or failed safety tests are hard stops regardless of aggregate score.

SIMPLE MARKET EXAMPLE

A candidate has moderate net performance but strong replication, low overlap, and robust operations. It may receive a small monitored pilot, while a higher-return idea with unresolved timing lineage remains blocked.

PYTHON / SYSTEM IMPLEMENTATION

Use a standard dossier, independent reviewers, documented exceptions, and stage-specific gates for research, paper trading, limited capital, and scale. Preserve the evidence available at each decision date.

CONFUSIONS TO AVOID

A weighted score can create false precision and allow tradeoffs that should be prohibited. Gates should not be weakened after seeing a desired candidate's result without a governed policy change.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The control plane binds approved scope, limits, artifact versions, owners, and expiry to the deployed strategy. Any material change triggers reevaluation at the appropriate gate.

RELATED CONCEPTS AND QUICK RECALL

Related: model approval, evidence dossier, pilot, policy exception. Recall task: Separate four hard-stop conditions from dimensions that can support a small monitored pilot despite uncertainty.

Research-to-production parity

BEGINNER

ONE-SENTENCE MEANING

Parity means the live system computes the same economic decision as the validated research under equivalent information, state, timing, and configuration. It requires shared logic or provably equivalent implementations plus continuous comparison.

MEMORY JOGGER

Same formula is insufficient if data arrives on a different clock, state resets differently, or execution uses another price. Compare complete decision records, not only feature values.

WHY IT MATTERS IN QUANT RESEARCH

Many strategies fail at handoff because batch research assumes finalized bars, unlimited history, or instantaneous orders. Parity testing exposes these differences before capital depends on them.

FORMULA INTUITION

A parity record compares sample key, data frontier, feature state, score, target position, order decision, and policy version. Tolerances must be justified by numerical or timing semantics rather than chosen after mismatches appear.

SIMPLE MARKET EXAMPLE

Research scores a close-based signal after the official bar is finalized, while live trading acts on a preliminary close. A two-minute finalization lag changes both feature value and executable price.

PYTHON / SYSTEM IMPLEMENTATION

Use shared transform artifacts, deterministic state machines, replay tests, batch-versus-stream fixtures, shadow scoring, and reconciliation dashboards. Log why every mismatch was expected or corrected.

CONFUSIONS TO AVOID

Matching aggregate P&L can hide offsetting sample-level errors. Online recomputation of rolling history may differ at warm-up, correction, restart, or calendar boundaries.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The deployment pipeline promotes immutable research artifacts into a serving package, while parity monitors gate traffic and maintain a rollback reference.

RELATED CONCEPTS AND QUICK RECALL

Related: batch-stream parity, shadow mode, replay, state machine. Recall task: Design a parity record for a close-to-open signal and identify three boundary conditions that need exact fixtures.

Pilot design and staged capital

BEGINNER

ONE-SENTENCE MEANING

A pilot exposes a validated strategy to controlled live conditions with limited capital, explicit learning objectives, stop rules, and no assumption that paper performance will transfer perfectly. It tests execution, data, monitoring, and organizational response as well as alpha.

MEMORY JOGGER

Start small enough to survive unknowns but large enough to observe the mechanism and operational path. Predefine what the pilot can teach and which result ends it.

WHY IT MATTERS IN QUANT RESEARCH

Live fills, borrow, latency, market impact, outages, and decision discipline cannot be fully inferred from simulation. Staging contains loss while gathering evidence about these uncertainties.

FORMULA INTUITION

Pilot limits can be based on worst credible loss, uncertainty-adjusted forecast, operational maturity, and detectability of failure. Scale increases only after minimum observations and control performance are met.

SIMPLE MARKET EXAMPLE

A daily cross-sectional strategy begins at five percent of target risk with per-name and turnover caps. Scale requires fill slippage within tolerance, signal calibration, stable health checks, and no unresolved parity mismatch.

PYTHON / SYSTEM IMPLEMENTATION

Declare objectives, metrics, minimum duration, capital and loss limits, data and execution health thresholds, manual authority, incident path, rollback package, and scale criteria before activation.

CONFUSIONS TO AVOID

A pilot is not permission to improvise around failed controls, and no-loss performance over a short calm period proves little. Scaling because P&L is positive consumes the experiment before learning objectives mature.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The control plane enforces pilot scope and expiry; monitoring collects decision-level evidence; governance approves scale, redesign, extension, or retirement from the predeclared report.

RELATED CONCEPTS AND QUICK RECALL

Related: paper trading, canary deployment, risk limit, scale gate. Recall task: Set learning objectives and stop rules for a low-turnover signal whose main uncertainty is live implementation cost.

Monitoring, decay, and retirement

BEGINNER

ONE-SENTENCE MEANING

Strategy monitoring tests whether inputs, mechanism proxies, calibration, costs, exposures, and outcomes remain inside the validated envelope. Retirement is the governed response when the mechanism, evidence, or operating context no longer supports capital use.

MEMORY JOGGER

Monitor the reason the strategy should work, not only its P&L. Separate market drawdown, model decay, data failure, execution deterioration, and unauthorized drift.

WHY IT MATTERS IN QUANT RESEARCH

A strategy can lose money while functioning as expected or remain profitable while its data pipeline is leaking. Mechanism and system monitors enable calibrated warning, degradation, stop, retraining, or retirement actions.

FORMULA INTUITION

Health combines data freshness and coverage, score distribution, calibration error, exposure drift, cost slippage, concentration, mechanism proxy, and drawdown relative to validated scenarios. Thresholds map to specific actions.

SIMPLE MARKET EXAMPLE

A liquidity-provision signal still predicts gross reversion, but spreads narrow and adverse selection rises, making net payoff negative. Monitoring attributes the change to execution economics rather than retraining the alpha model blindly.

PYTHON / SYSTEM IMPLEMENTATION

Maintain warning, degraded, and hard-stop states with owners, runbooks, conservative fallbacks, and immutable incident evidence. Use sequential or rolling tests carefully and review thresholds on a fixed cadence outside incidents.

CONFUSIONS TO AVOID

Automatic retraining can absorb corrupt data or destroy a stable mechanism, while stopping solely on recent P&L can lock in noise. Retirement should preserve artifacts and lessons rather than erase the failed history.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Monitoring connects live decisions to research lineage, risk controls, incident response, and the idea registry. Retired strategies remain searchable as negative knowledge and may be reconsidered only under a new hypothesis.

RELATED CONCEPTS AND QUICK RECALL

Related: model drift, calibration, degraded mode, retirement policy. Recall task: Define warning, degraded, and stop triggers that distinguish data failure, cost deterioration, and genuine signal decay.

SYSTEM DIAGRAM

Research funnel and evidence states

BEGINNER

HOW TO READ IT

The funnel reduces freedom while evidence and operational responsibility increase. Rejections remain durable knowledge; only candidates with a traceable contract move into capital-bearing states.

OBSERVATION	Market behavior or research source with context and owner
IDEA	Mechanism, decision, required data and counterarguments
HYPOTHESIS	Locked direction, endpoint, search family and stop rule
CANDIDATE	Causal sample, diagnostics, costs and temporal evidence
REPLICATED	Independent boundary, negative controls and red-team review
PILOT	Parity, limits, live learning objectives and rollback
SCALED	Approved scope, monitors, change control and retirement

GOVERNED EXITS

At any state the candidate may be rejected, parked pending data, merged with a duplicate, limited to a narrow regime, or returned for redesign. Preserve the reason, evidence available at the time, and conditions that would justify reconsideration.

BLOCK DIAGRAM

Anatomy of a researchable signal

BEGINNER

DESIGN QUESTION

A signal is a chain of semantic contracts. Changing any clock, sample, transform, calibration, or action creates a different hypothesis even when the indicator name stays the same.

1 EVIDENCE

Raw and canonical data, source vintage, availability, units, quality and stable identity

2 SAMPLE

Point-in-time universe, decision clock, observation window, exclusions and target interval

3 FEATURE

Economic measurement with declared grain, lookback, aggregation and missing-state policy

4 MAPPING

Direction, center, scale, clipping, rank or calibration fitted only from training evidence

5 FORECAST

Expected outcome and uncertainty in consistent horizon, return, risk and cost units

6 ACTION

Position, threshold, rebalance, constraints, execution path and fallback under poor health

AUDIT TEST

Select one live score and reconstruct each block from immutable artifacts. If the decision cannot be reproduced from the original information frontier, the object is a research result without a trustworthy system contract.

DESIGN SCORECARD

Hypothesis quality scorecard

BEGINNER

DECISION

The action, information cutoff, universe, horizon, benchmark, constraints, cost model, and minimum useful effect are explicit enough for another researcher to implement.

MECHANISM

Actors, incentives, transmission path, expected direction, decay, state dependence, capacity, and two disconfirming observations are named before result access.

MEASUREMENT

Feature, label, sample grain, units, availability clocks, revisions, missing states, and peer groups represent the intended economics without future information.

PRIMARY EVIDENCE

Estimator, split policy, primary endpoint, uncertainty, multiplicity family, baseline, and stop rule are locked; secondary results cannot silently replace them.

IMPLEMENTATION

Execution clock, delay, turnover, spread, impact, borrow, liquidity, position limits, and unwind path are connected to the forecast horizon.

FALSIFICATION

Negative controls, placebo outcomes, alternative explanations, parameter neighborhood, concentration, and independent replication have predicted interpretations.

GOVERNANCE

Idea source, search history, owners, immutable parents, reviewer, permitted scope, expiry, change policy, monitor thresholds, and retirement path are recorded.

REJECT WHEN

The effect requires post-cutoff data, depends on an undocumented search, cannot survive its own costs, fails a mechanism-specific control, or cannot be reproduced from pinned evidence.

CONCEPT MAP

Causal map for an attention signal

BEGINNER

QUESTION

Does filing complexity create slow information processing, or does a complexity proxy merely identify distressed, illiquid firms? The map turns these explanations into different controls and falsification tests.

TEST DESIGN

Measure processing delay, control prior distress without conditioning on post-filing returns, stratify liquidity, test a pre-event placebo, and estimate executable payoff separately from predictive drift.

WORKED CALCULATION

Worked signal mapping and calibration

BEGINNER

SCENARIO

A valuation spread is expected to mean revert. Training data estimate a robust center of 0.40 and scale of 0.25. Larger positive spreads imply lower future returns, and scores are capped at three scale units.

FORMULA BLOCK

z = (x - 0.40) / 0.25; clipped z = max(-3, min(3, z)); signal s = -clipped z. The negative sign encodes mean reversion, while clipping limits the influence of sparse or unreliable extremes.

NAME	RAW X	Z	SCORE	ACTION	INTERPRETATION
A	0.10	-1.20	+1.20	**long**	below center; expected rebound
B	0.40	0.00	0.00	**flat**	at robust center
C	0.90	+2.00	-2.00	**short**	rich versus reference
D	1.60	+4.80	-3.00	**short cap**	tail capped; inspect distress
E	missing	n/a	n/a	**ineligible**	do not replace with zero

VALIDATION STEPS

Fit center, scale, cap, and sign only on training evidence. Plot future outcomes by score bin with counts and intervals, compare unclipped and ranked variants, test the tail separately, and verify that live values use the identical fitted artifact.

CONFUSION TO AVOID

A capped score is not a return forecast until calibration maps score units to expected payoff and uncertainty. Position size must additionally reflect risk, covariance, costs, constraints, and health state.

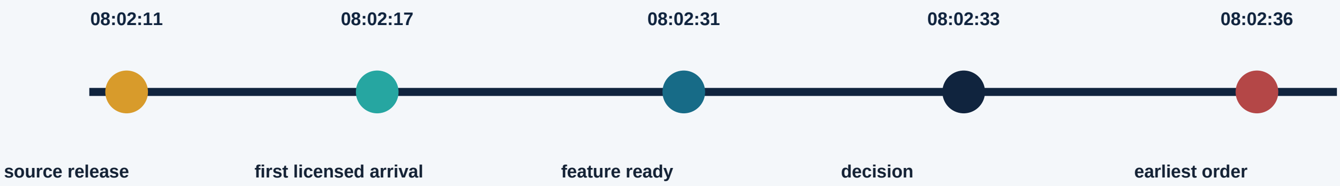
TIMELINE

Information, decision, and execution clocks

BEGINNER

SCENARIO

A filing is released at 08:02:11, reaches the licensed feed at 08:02:17, finishes parsing at 08:02:31, generates a decision at 08:02:33, and may first trade after risk checks at 08:02:36.



FEATURE CUTOFF

Only records with licensed arrival and required processing completed by 08:02:31 are eligible. The public release time alone does not prove the strategy could consume the content.

TARGET START

The research label must begin at an executable price at or after 08:02:36, not at the 08:02:11 market state. The six-to-twenty-five second gap may contain most of the price response.

REVISION POLICY

Later corrected timestamps or document text create new data vintages. Historical research pins first-seen evidence and separately measures the effect of revisions.

LATENCY STRESS

Replay slower parsing, risk queues, and order routing. A viable event signal should report payoff remaining at multiple realistic delays, not only at the best timestamp.

CLOCK INEQUALITY

For every sample: maximum availability time of required inputs <= feature-ready cutoff < decision time <= order submission time <= target entry time. Store the actual values rather than assuming a daily row satisfies them.

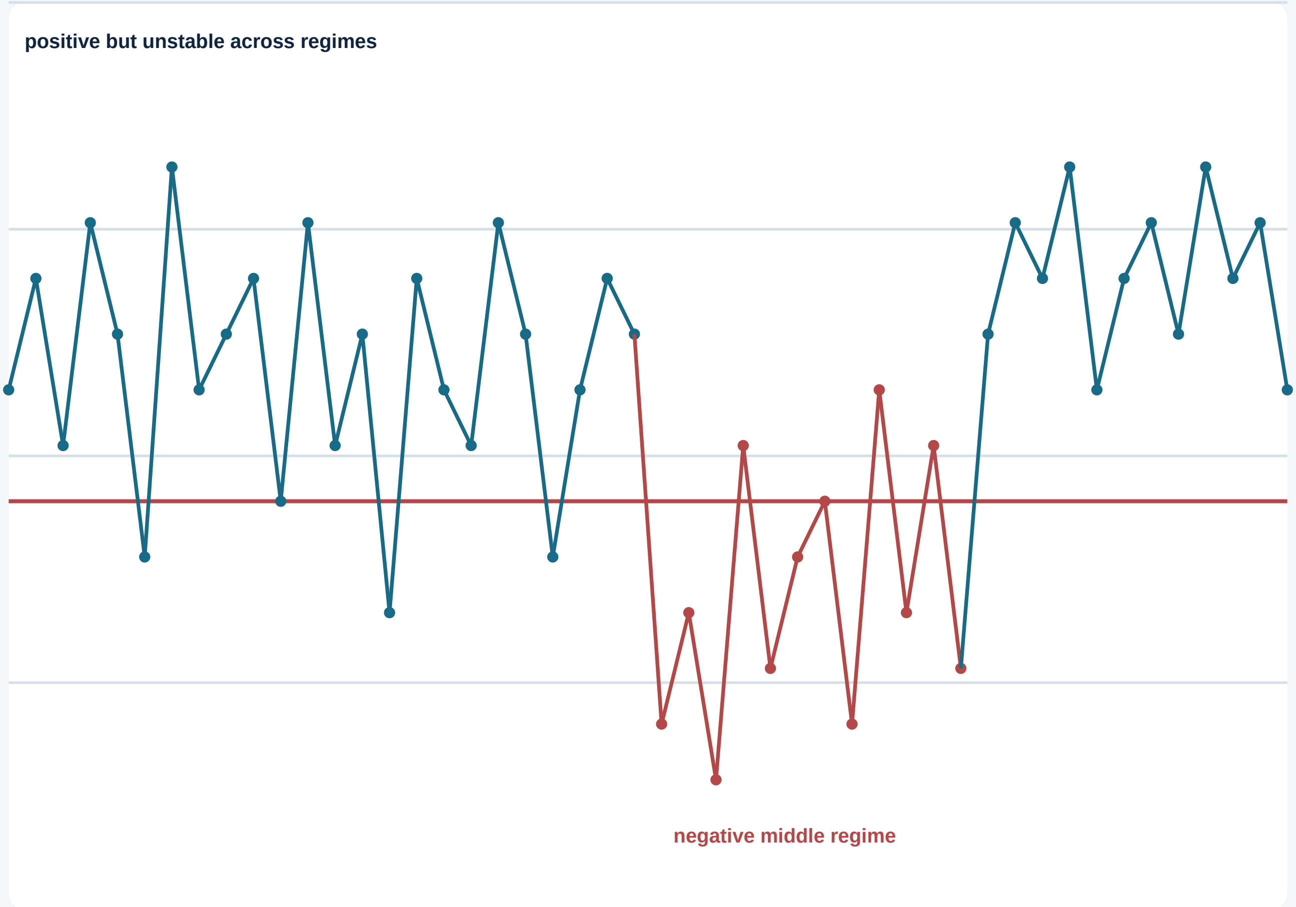
ANNOTATED CHART

Read an IC distribution, not one mean

BEGINNER

SYNTHETIC DIAGNOSTIC

Forty-eight monthly rank-IC observations have a modest positive mean, one negative regime, and unequal instrument counts. The chart is educational; values are not empirical claims.



INTERPRETATION

Report mean IC with dependence-aware uncertainty, median, hit rate, autocorrelation, per-period eligible counts, and contribution by regime. Investigate whether the negative block reflects a forecast failure, changed mechanism, or data and universe issue.

NEXT TESTS

Recompute with fixed eligibility and alternative weights, compare Pearson and rank estimators, leave out each era, test baseline controls, and translate forecast ordering into net quantile portfolios. Do not infer portfolio Sharpe from IC alone.

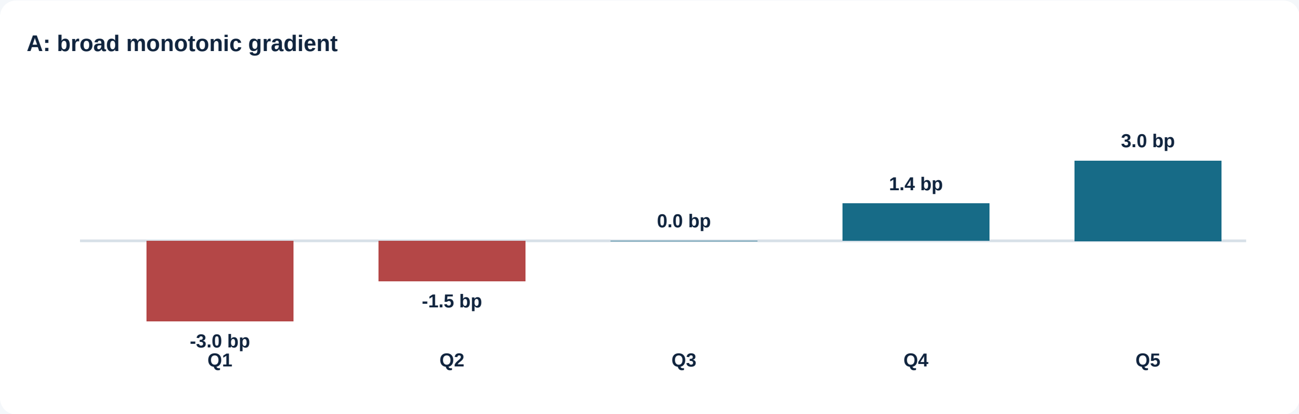
COMPARISON CHART

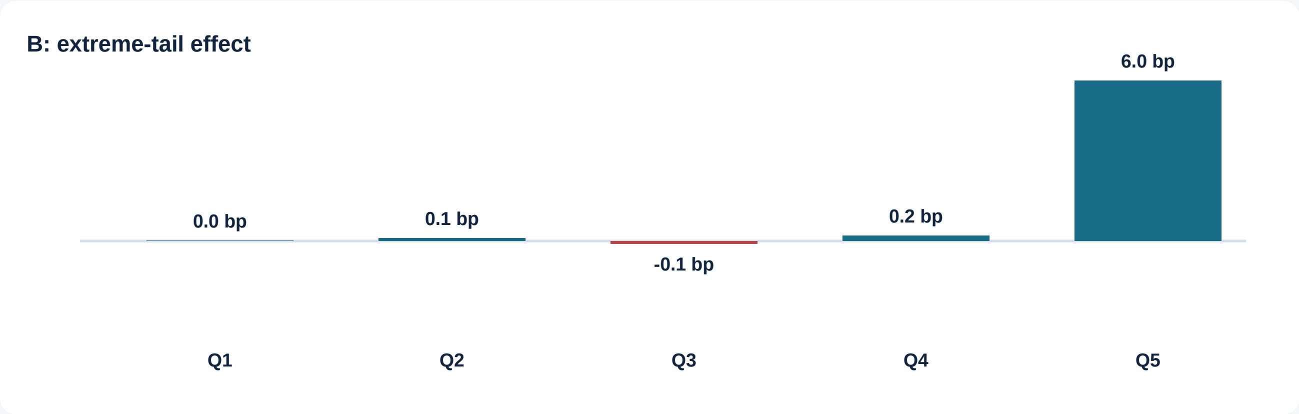
Quantile profile: gradient versus tail

BEGINNER

TWO SYNTHETIC SIGNALS

Signal A shows an ordered payoff gradient. Signal B earns only in the extreme top bucket. Both have the same top-minus-bottom spread, but they imply different mechanisms, mappings, and capacity.





DIFFERENT CONCLUSIONS

Signal A supports a continuous ranking, subject to cost and exposure checks. Signal B may support a threshold trade, but first requires tail sample counts, outlier and distress review, borrow and liquidity analysis, fixed breakpoints, and independent replication.

COMPARISON RULE

Always publish all buckets, counts, dispersion, turnover, factor exposures, and net returns. A top-minus-bottom number alone discards the response shape that determines implementation.

HEATMAP

Parameter surface and robustness plateau

BEGINNER

SYNTHETIC SURFACE

Cells show an out-of-sample quality score across lookback and holding horizon. The central region is intentionally broad; selection should favor a simple representative, not the numerically highest neighboring cell.

	1 day	3 days	5 days	10 days	20 days	40 days
10 days	WEAK	WEAK	GOOD	WEAK	FAIL	FAIL
20 days	WEAK	GOOD	PLATEAU	GOOD	WEAK	FAIL
40 days	GOOD	PLATEAU	PLATEAU	PLATEAU	GOOD	WEAK
60 days	GOOD	PLATEAU	PLATEAU	PLATEAU	GOOD	WEAK
90 days	WEAK	GOOD	PLATEAU	GOOD	GOOD	WEAK
120 days	FAIL	WEAK	GOOD	GOOD	WEAK	FAIL

SELECTION

Choose a forty- or sixty-day lookback with a five- or ten-day horizon only if the mechanism supports that time scale. Lock the selection rule inside development data, then evaluate the chosen cell once on sealed evidence.

CAUTION

Neighboring cells share observations, so a plateau is not independent replication. Preserve every tested cell for search accounting, stress grid boundaries, and reject a surface whose favorable region appears only after expanding the search.

WORKED CALCULATION

Worked multiple-testing calculation

BEGINNER

SCENARIO

A project tests 20 feature definitions, 5 lookbacks, 4 horizons, 3 universes, and 2 neutralization choices. If every combination is inspected, the visible search family contains 2,400 specifications before manual chart choices.

SEARCH COUNT

$m = 20 \times 5 \times 4 \times 3 \times 2 = 2{,}400$. Under independent true nulls and a nominal 5 percent threshold, the expected number of false positives is $2{,}400 \times 0.05 = 120$. Dependence changes the distribution, not the need to account for the search.

BONFERRONI FAMILY-WISE THRESHOLD

$0.05 / 2{,}400 = 0.0000208$. This is conservative, especially under correlated tests, but it shows how little a nominal p-value of 0.01 means after a large search.

FALSE-DISCOVERY CONTROL

Order all p-values and compare $p_{(i)}$ with $q*i/m$ under the chosen procedure and assumptions. Report the complete family and the expected false-discovery proportion, not only accepted names.

SELECTION BIAS

The best observed effect is upward biased even when it passes a threshold. Use nested selection, shrink expected performance, and preserve a sealed forward period for the exact chosen specification.

HUMAN SEARCH

Viewed charts, changed filters, excluded eras, and revised narratives also consume evidence. Log material decisions and increase replication burden when their effective count cannot be estimated precisely.

INDEPENDENT CONFIRMATION

Freeze code, data policy, transform, horizon, universe, and evaluation before opening later data or a new market. Any post-result change begins another exploratory branch.

DECISION

The top result is a candidate, not a discovery. Its evidence grade depends on the whole search, honest forward confirmation, mechanism tests, implementation economics, and independent challenge.

STABILITY MAP

Regime stability and concentration map

BEGINNER

SYNTHETIC EVIDENCE

A candidate is evaluated across market states and liquidity cohorts. Each cell combines signed net effect and minimum sample support; the map reveals where the mechanism is broad, conditional, or contradicted.

	Most liquid	Liquid	Mid liquidity	Less liquid	Small / sparse
Quiet uptrend	+ STABLE	+ STABLE	+	WEAK	N/A
Volatile uptrend	+	+	WEAK	-	N/A
Quiet decline	WEAK	WEAK	0	-	N/A
Volatile decline	-	-	-	- UNSTABLE	N/A
Funding stress	-	-	-	N/A	N/A
Recovery	+	+ STABLE	+ STABLE	+	WEAK

INTERPRETATION

The candidate behaves like a risk-sensitive effect rather than a universal alpha: it weakens in declines and fails during funding stress. Decide whether that pattern is predicted compensation, an unacceptable portfolio overlap, or evidence against the mechanism.

EVIDENCE DISCIPLINE

Preserve counts, capital, turnover, cost, and intervals for every cell. Do not delete adverse states or promote the favorable recovery row to a new strategy without registering a conditional hypothesis and obtaining independent evidence.

COST WATERFALL

Gross forecast to executable payoff

BEGINNER

SCENARIO

A short-horizon candidate forecasts 12 basis points before execution. The waterfall applies measured delay decay and conservative implementation costs for a representative trade size.

12.0 bp	**Gross forecast**	expected move at feature-ready time
-2.5 bp	**Decision and routing delay**	price response before earliest executable order
-1.2 bp	**Half spread**	crossing or adverse passive-fill expectation
-0.3 bp	**Fees and rebates**	venue and broker schedule
-2.0 bp	**Market impact**	size and participation under normal conditions
-0.4 bp	**Borrow and financing**	holding-period estimate with availability rule
-1.0 bp	**Risk reserve**	uncertainty and stressed slippage allowance
+4.6 bp	**Expected net**	remaining decision value before portfolio overlap

STRESS QUESTION

At wider spreads, slower routing, and doubled impact, expected net becomes negative. A pilot must cap size and stop when measured cost or latency crosses the predeclared break-even surface.

DIAGNOSTIC TABLE

Signal research failure diagnostics

BEGINNER

SYMPTOM	LIKELY CAUSE	FIRST EVIDENCE	CONTROL ACTION
Huge metric, weak story	Leakage or broad search	Clock lineage and family history	Replay first-seen data; seal new test
Only best bucket works	Tail outlier or threshold	Counts, borrow, distress, influence	Fix breakpoints; replicate tail
Peak at one parameter	Specification luck	Full surface and boundary grid	Choose plateau or reject
IC positive, P&L weak	Costs, covariance, constraints	Decay and gross-to-net waterfall	Align horizon and execution
Strong one era	Regime or data-process dependence	Contribution and mechanism proxies	Register conditional use or limit
Baseline removes effect	Duplicate known exposure	Paired incremental forecast	Reframe novelty or combine
Placebo also works	Timing or confounding defect	Pre-event and unaffected cohorts	Block family; audit pipeline
Live and research differ	State or clock parity failure	Decision-level reconciliation	Stop, replay, and roll back
Costs drift upward	Crowding or liquidity change	Fill, spread, impact and participation	Reduce scale or retire

TRIAGE RULE

Find the earliest boundary where information, sample, search, economics, or live behavior diverges from the contract. Downstream P&L is impact evidence, not permission to patch the final output in place.

EVIDENCE DOSSIER

Evidence grade and pilot scope

BEGINNER

MECHANISM - STRONG

Actors, incentives, transmission, decay, capacity, and falsifiers produced distinct predictions that were observed without material contradiction.

DATA INTEGRITY - STRONG

Point-in-time universe, first-seen timestamps, revisions, stable identity, labels, transforms, lineage, and quality checks are replayable from immutable parents.

STATISTICAL EVIDENCE - MODERATE

Primary forward tests pass with corrected evidence, but intervals remain wide and the search family was larger than ideal. Performance is not treated as precisely known.

ROBUSTNESS - MODERATE

Parameter neighborhood and most cohorts are coherent; one adverse regime and limited external-market replication constrain confidence and permitted scope.

IMPLEMENTATION - STRONG

Delay, spread, impact, borrow, turnover, capacity, unwind, parity, fallback, and stress behavior are measured at the intended initial size.

INDEPENDENT CHALLENGE - STRONG

Negative controls pass, red-team findings are closed or accepted explicitly, and the replication package reproduces sample-level decisions.

DECISION

Permit a time-limited pilot at reduced risk with strict latency, cost, regime, concentration, and parity thresholds. No scale increase until minimum live observations and external replication complete.

HARD STOPS

Any future-data dependency, unresolved license restriction, unreproducible artifact, failed control, unsafe rollback, or material live mismatch blocks the pilot regardless of aggregate grade.

PROMOTION GATE

Strategy research promotion checklist

BEGINNER

DECISION CONTRACT

Action, universe, clock, horizon, benchmark, minimum useful effect, constraints, costs, and rejection rules are frozen and linked to an owner.

MECHANISM AND ALTERNATIVES

Actors, incentives, causal map, proxy exposures, state dependence, decay, capacity, counterarguments, and falsification tests are documented.

SEARCH ACCOUNTING

All material transforms, lags, cohorts, models, charts, deviations, and abandoned branches are attached to a search family with corrected evidence.

CAUSAL SAMPLE

Point-in-time eligibility, availability, first-seen vintages, labels, purging, missing states, and fitted transformations pass replay and future-perturbation tests.

SIGNAL DIAGNOSTICS

Support, response shape, IC, ordered portfolios, baseline increment, concentration, decay, turnover, cost, parameter surface, and exposure attribution are acceptable.

INDEPENDENT EVIDENCE

Temporal holdout, negative controls, placebos, portability boundary, independent replication, and adversarial findings support the declared interpretation.

IMPLEMENTATION

Research-to-live parity, execution simulation, borrow, impact, capacity, limits, fallback, rollback, entitlements, and incident procedures are verified.

PILOT AND MONITORING

Learning objectives, minimum duration, scale gates, health states, mechanism metrics, cost thresholds, ownership, expiry, and retirement rules are enforced.

FINAL STANDARD

No single Sharpe, p-value, or narrative compensates for a critical failure in clocks, lineage, search honesty, net economics, safety, or reproducibility.

FINAL REVIEW

Final active recall and system index

BEGINNER

FRAME

Translate an observation into a capital decision with universe, clock, action, horizon, benchmark, constraints, costs, useful effect, and rejection rule.

EXPLAIN

Name the actor, incentive or constraint, transmission path, response, decay, capacity, competing explanations, and observations that would falsify the mechanism.

LOCK

Predeclare direction, measurement, sample, primary endpoint, estimator, split, baseline, multiplicity family, and stop rule before confirmatory evidence.

BUILD

Create a point-in-time sample with stable identity, availability, first-seen data, causal labels, fold-fitted transforms, missing-state policy, and immutable lineage.

DIAGNOSE

Read support, score mapping, IC distribution, complete quantile profile, incremental value, decay, turnover, costs, sensitivity, and contribution concentration.

CHALLENGE

Use temporal folds, negative controls, placebos, regime maps, parameter plateaus, independent replication, and red-team attacks matched to the mechanism.

VALUE

Translate forecast into marginal net portfolio value after covariance, constraints, capacity, tail risk, operational cost, and uncertainty.

GOVERN

Record every material trial, grade evidence with hard stops, prove live parity, stage capital, enforce scope, and preserve independent approval.

MONITOR

Watch data, mechanism proxies, score distribution, calibration, exposure, cost, execution, incidents, and decision sensitivity with warning, degraded, and stop actions.

RETIRE

Stop or redesign when the mechanism, evidence, economics, or controls leave the validated envelope; preserve the full history as reusable negative knowledge.